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Determining the effects of disposable income on the quality of provided mental health services

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Introduction

Income is an indicator for socioeconomic status

- The likelihood of mental health issues is associated with income and socioeconomic status (Nickel, 2020)
- Obstacles associated with lower income lead to difficult in accessing treatment for mental health difficulties (Hodgkinson et al., 2016)

Risk factors of mental health include:

- Financial strain
- Housing insecurity
- Discrimination

All are linked to a low socioeconomic status (Braveman & Gottlieb, 2014)

Objective:

- Investigate the effects that disposable income has on the quality of mental health services provided to a community

Methods

Data Collection

- NHS Digital: Mental Health Services Dataset
- NHS: Patient experience of community mental health services
- NHS: Mental Health and Learning Disabilities Data Set

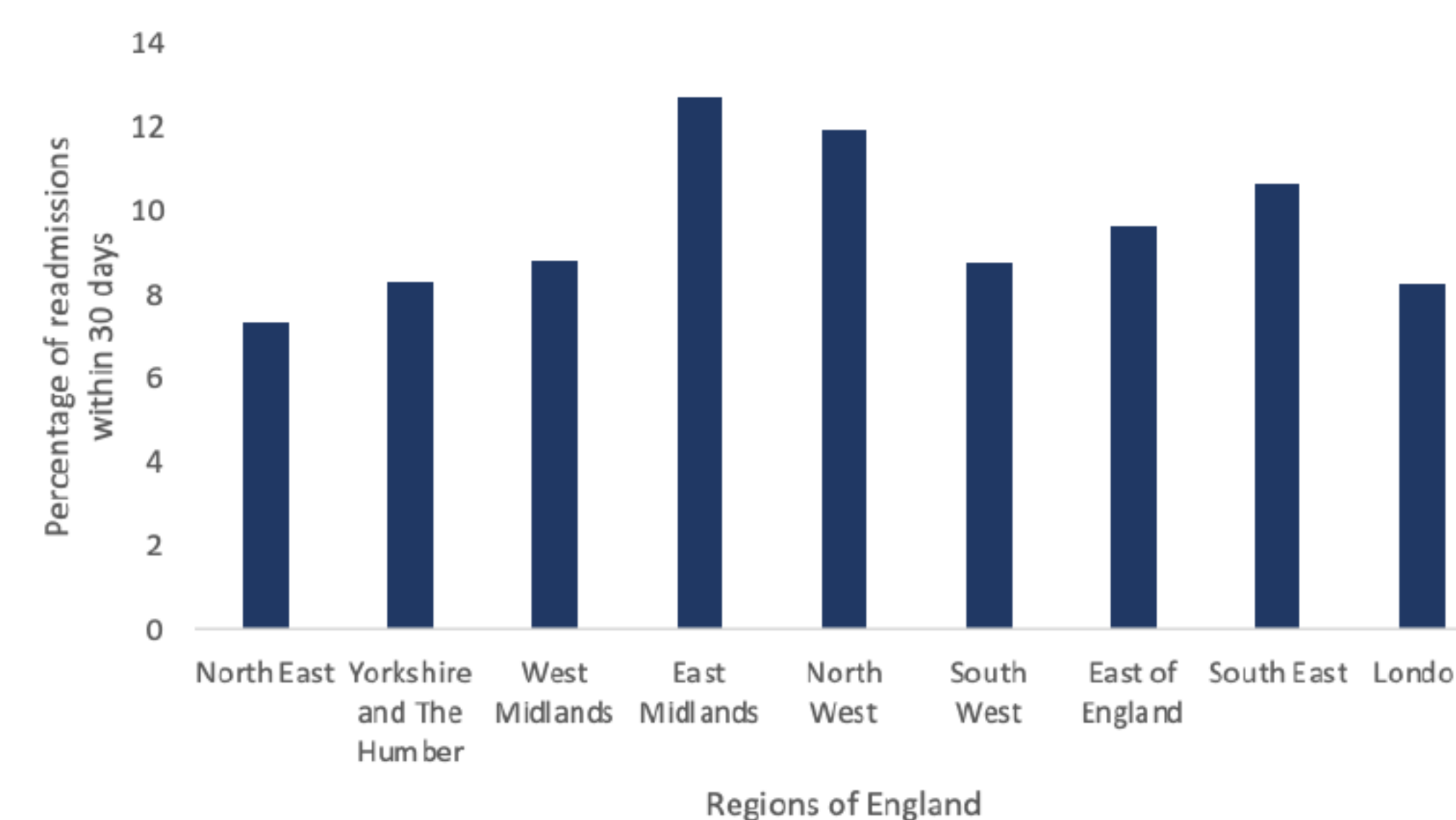
Data Wrangling

- Done using Excel
- Removed health care providers that provided services to multiple regions

Data Analysis

- Done using R Studio
- Two-sided t-test used to compare values of normally distributed data
- Kruskal-Wallis test used to compare values of not normally distributed data
- Bar graphs created using Excel

Results



Percentage of readmission to mental health services

Indicates that regions with moderate levels of income have the highest levels of readmission.



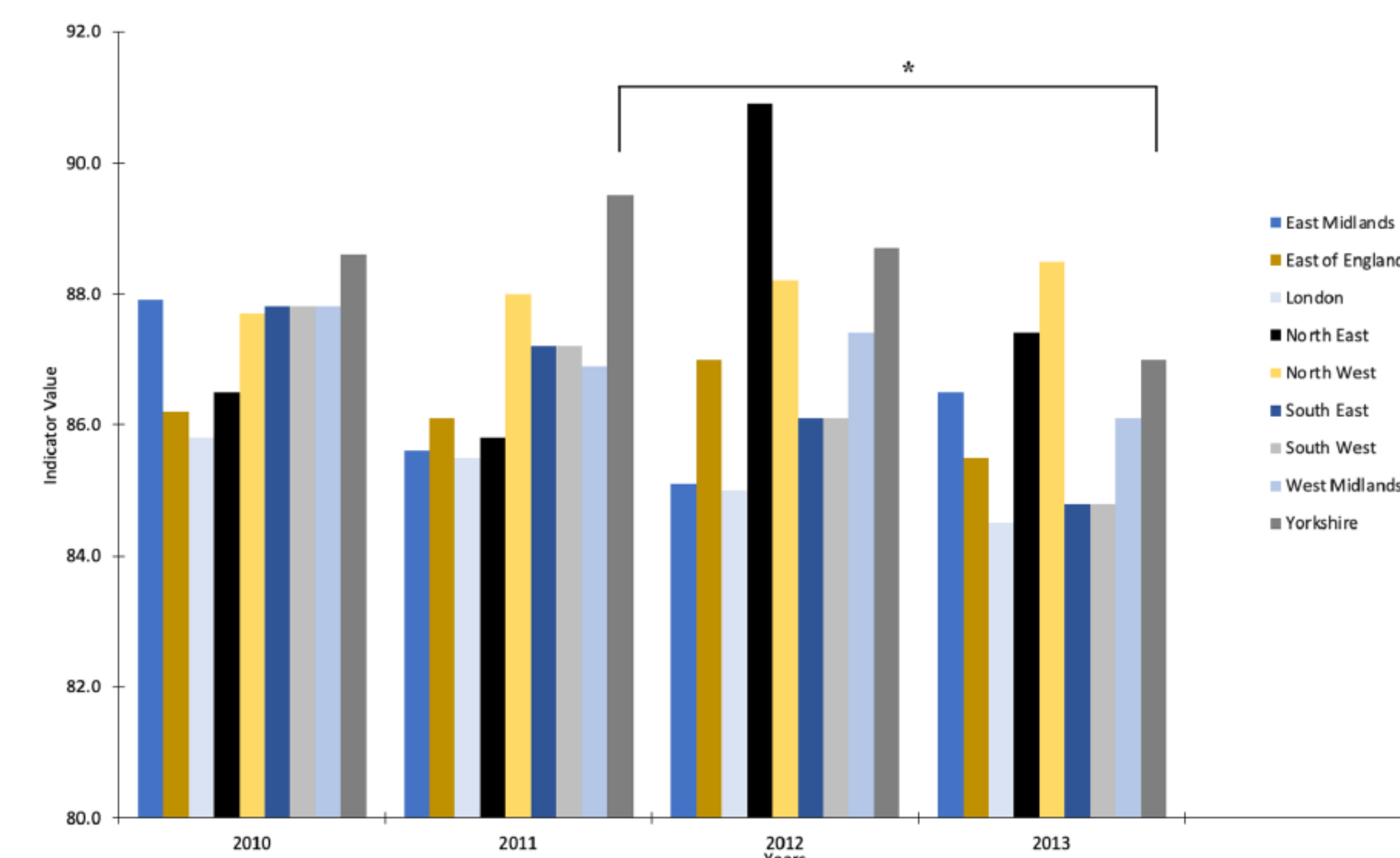
Percentage of population in contact with mental health services

Indicates that regions with moderate levels of income have a lower percentage of adults in contact with mental health services



Satisfaction with health care provider between regions with varying GDHI

Significant difference between indicator values for areas of with lower income relative to other regions. Suggests that populations with lower income are more satisfied with their service



Change in satisfaction with health care provider from 2010-2013

Indicator value tends to remain static throughout the years for all levels of income.

Discussion

Populations with moderate levels of income:

- Lower percentage of adults in contact with mental health services
- Highest levels of readmission to mental health services after 30 days of treatment

Populations with lower levels of income:

- Highest satisfaction with health care providers

Overall, no significant change in the satisfaction with service quality was reported over the years.

Future Considerations

- Using simple linear regression to determine if there is a correlation between income level and the variables that were tested
- Looking into other countries where mental health care is paid out of pocket

References

- Braveman, P., & Gottlieb, L. (2014). The Social Determinants of Health: It's Time to Consider the Causes of the Causes. *Public Health Reports*, 129(1), 13. <https://journals.sagepub.com/doi/epdf/10.1177/003335491412915206>
- Hodgkinson, S., Godoy, L., Beers, L. S., & Lewin, A. (2016). Improving Mental Health Access for Low-Income Children and Families in the Primary Care Setting. *NCBJ*. DOI: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5192088/>
- Nickel, S., & von dem Knesebeck, O. (2020). Do multiple community-based interventions on health promotion tackle health inequalities? *International Journal for Equity in Health*, 19(1). <https://doi.org/10.1186/s12939-020-01271-8>